Bias terms and decision boundaries

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One readable score and boundary record.

Trace arithmetic,plot named coordinates and explain the evidence limit.

Keep one habit: W1–W3/W5 → selected new boundary model → project graph → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Dot products and weighted sums. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Use the fictional parcel schema [width in cm, height in cm]. Our model computes z = 2 times width - height - 3. The weights are [2, -1]; the bias b is -3. Compute the weighted sum first, then add the bias once.

  The result is an affine score: a weighted sum plus a constant offset.

- Now declare a classification rule: predict class 1 when z is at least 0, and class 0 when z is below 0. Equality belongs to class 1. These are invented categories for arithmetic practice, not a real parcel decision.

  The weights and bias are supplied, so this is a forward calculation, not **training**.

- A bias is a mathematical parameter. It is not the same use of the word as unfair treatment; numerical offset alone does not establish whether a system is fair.

Dot products and weighted sums

> **My remembered idea, example and correction**

Lesson reference for workbook practice

Input [width, height]	Score z	Prediction
[4, 2]	8 - 2 - 3 = 3	1
[2, 4]	4 - 4 - 3 = -3	0
[3, 3]	6 - 3 - 3 = 0	1 at equality

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main model z=2 width−height−3, class 1 when z≥0. Weights[2, −1], bias−3, inputs in cm. A[4, 2]z3/class 1, B[2, 4]z−3/class 0, C[3, 3]z0/class 1. Boundary height=2 width−3 through(2, 1), (3, 3), (4, 5); class 1on/below line with width horizontal. W2[4, 4]z1/class 1, [4, 6]z−1/class 0. Change b−3→−1 adds 2to every score and gives parallel boundary height=2 width−1.

  Separate pictured A[2, 1]0→2, B[1, 1]−2→0.≥versus>changes equality assignment, not the zero line. Supplied parameters are not training or proof of improvement; missing/incorrect units violate input assumptions.

- Use z = 2 times width - height - 3 at [5, 4]. Give the weighted sum, bias, final score and **prediction** under z >= 0.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main model z=2 width−height−3, class 1 when z≥0. Weights[2, −1], bias−3, inputs in cm. A[4, 2]z3/class 1, B[2, 4]z−3/class 0, C[3, 3]z0/class 1. Boundary height=2 width−3 through(2, 1), (3, 3), (4, 5); class 1on/below line with width horizontal. W2[4, 4]z1/class 1, [4, 6]z−1/class 0. Change b−3→−1 adds 2to every score and gives parallel boundary height=2 width−1.

  Separate pictured A[2, 1]0→2, B[1, 1]−2→0.≥versus>changes equality assignment, not the zero line. Supplied parameters are not training or proof of improvement; missing/incorrect units violate input assumptions.

- For the worked model, find boundary points at width 2 and width 4. Test [4, 4] and [4, 6] to identify the class on each side.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main model z=2 width−height−3, class 1 when z≥0. Weights[2, −1], bias−3, inputs in cm. A[4, 2]z3/class 1, B[2, 4]z−3/class 0, C[3, 3]z0/class 1. Boundary height=2 width−3 through(2, 1), (3, 3), (4, 5); class 1on/below line with width horizontal. W2[4, 4]z1/class 1, [4, 6]z−1/class 0. Change b−3→−1 adds 2to every score and gives parallel boundary height=2 width−1.

  Separate pictured A[2, 1]0→2, B[1, 1]−2→0.≥versus>changes equality assignment, not the zero line. Supplied parameters are not training or proof of improvement; missing/incorrect units violate input assumptions.

- At [3, 3], compare **prediction** under z >= 0 with **prediction** under z > 0. Does changing this comparison move the zero line?

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main model z=2 width–height–3, class 1 when z≥0. Weights[2, –1], bias–3, inputs in cm. A[4, 2]z3/class 1, B[2, 4]z–3/class 0, C[3, 3]z0/class 1. Boundary height=2 width–3 through(2, 1), (3, 3), (4, 5); class 1on/below line with width horizontal. W2[4, 4]z1/class 1, [4, 6]z–1/class 0. Change b–3→–1 adds 2to every score and gives parallel boundary height=2 width–1.

  Separate pictured A[2, 1]0→2, B[1, 1]–2→0.≥versus>changes equality assignment, not the zero line. Supplied parameters are not training or proof of improvement; missing/incorrect units violate input assumptions.

- A teammate says that raising b must improve the model because all scores rise. What information is needed before accepting an improvement claim?

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A new model has w = [-1, 2], b = -2, **input**s [u, v] and class 1 when z >= 0. Calculate scores and classes for [2, 1], [2, 2], [2, 3]. Write the boundary as v in terms of u. Then change only b to 0 and recompute [2, 1].

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Analyse a new model independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Add a labelled **bias term**s and decision boundaries **evidence** sheet to Image-pattern Network Investigation. Show how you can calculate a weighted score with a **bias term**.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Plot and label the boundary for **Bias term**s and decision boundaries. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Bias term**s and decision boundaries. Proposed explanation: "Add bias to each contribution". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A new model has w = [-1, 2], b = -2, **input**s [u, v] and class 1 when z >= 0. Calculate scores and classes for [2, 1], [2, 2], [2, 3]. Write the boundary as v in terms of u. Then change only b to 0 and recompute [2, 1].

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

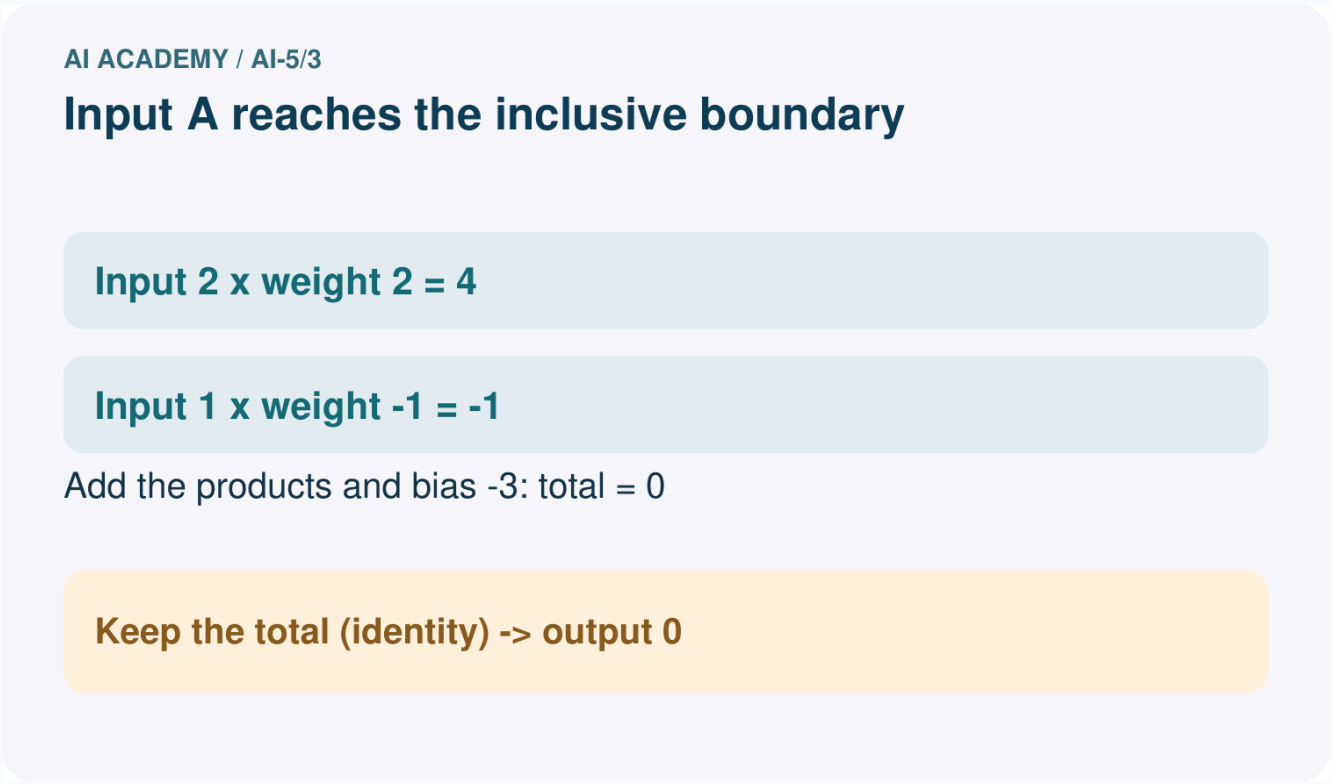
> Keep this task's inputs,rule and response together.

Complete task and evidence

- Move a toy boundary with a bias while keeping its **input** weights fixed.
- Score z=2u-v-3; predict class 1 when z>=0. **Input**sA=[2, 1], B=[1, 1].
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Is B positive under the original or changed model? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/3

Bias terms and decision boundaries

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Trace input A

A score=4-1-3=0, so class 1 at the declared inclusive boundary.



2. Trace input B

B score=2-1-3=-2, so class 0.



3. Read boundary

The boundary is v=2u-3; the bias sets its position.

CHANGE ONE THING

Change bias from -3 to-1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Move a toy boundary with a bias while keeping its **input** weights fixed.
- Which labelled step matches this detail? The boundary is v=2u-3; the bias sets its position.
- 1. Trace **input** A
- 2. Read boundary
- 3. Trace **input** B
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Read boundary The boundary is v=2u-3; the bias sets its position. The bias adds the same offset after the **input** contributions.

It moves the dividing line without changing the weights that set the line's orientation. Class 1 includes score zero under our stated rule.

That is why B changes class when the bias rises by two: its score moves from minus two onto the included boundary.

Changed-case question and optional hint

- Changed-case practice: Change bias from -3 to-1. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. A score becomes 2 and B score 0; both predict class 1.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: A score becomes 2 and B score 0; both predict class 1. Increasing bias by2 adds 2 to every score without changing the weights.

Repair and retain the record

Review Dot products and weighted sums (ai-5/2). Goal: Calculate a dot product by matching feature positions. Short practice: Which connection explains why more right brightness lowers this score?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Bias terms and decision boundaries: the case

AI ACADEMY / AI-5/3 · 01 · LOOK AT THE FACTS

Bias terms and decision boundaries: the case

The situation

Move a toy boundary with a bias while keeping its input weights fixed.

Supplied evidence

Score z=2u-v-3; predict class 1 when z>=0. InputsA=[2, 1], B=[1, 1].
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain bias terms and decision boundaries using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

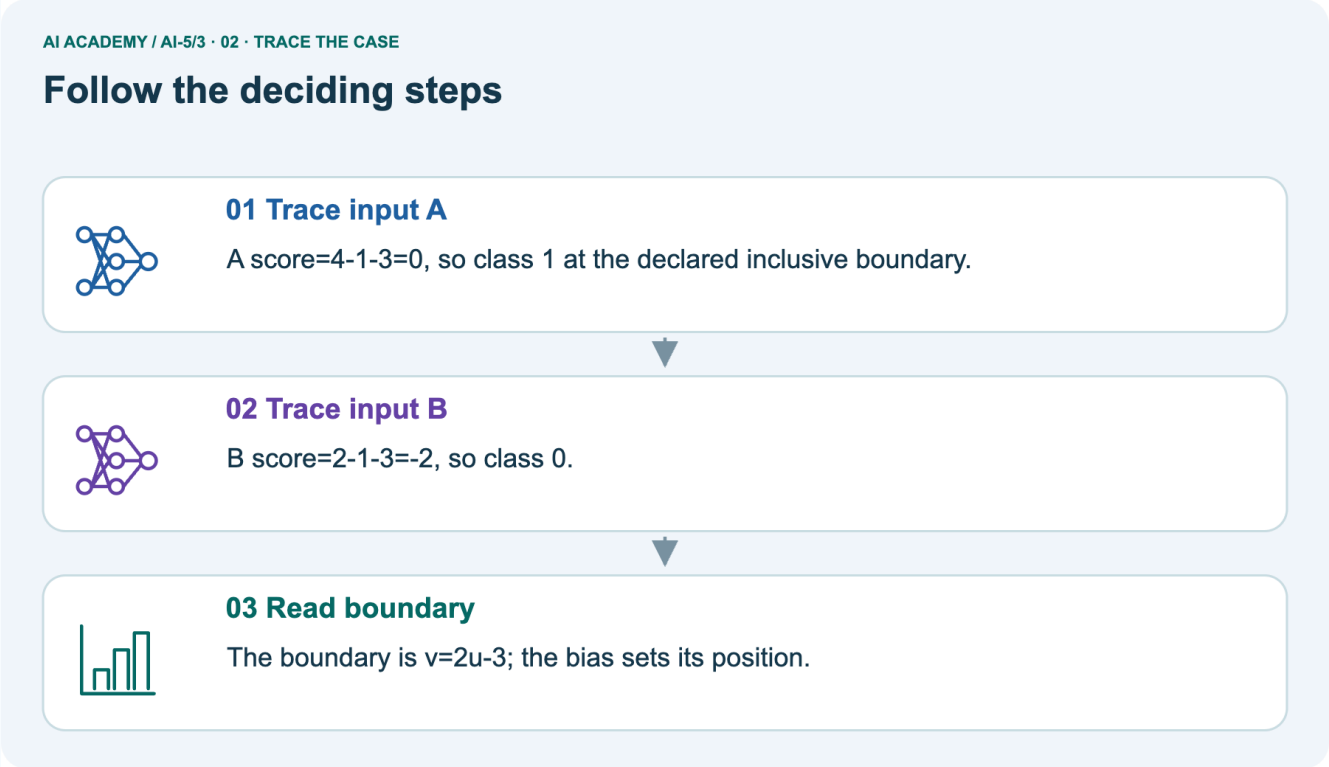
Move a toy boundary with a bias while keeping its **input** weights fixed. Supplied facts: Score z=2u-v-3; predict class 1 when z>=0. **Input**sA=[2, 1], B=[1, 1].

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



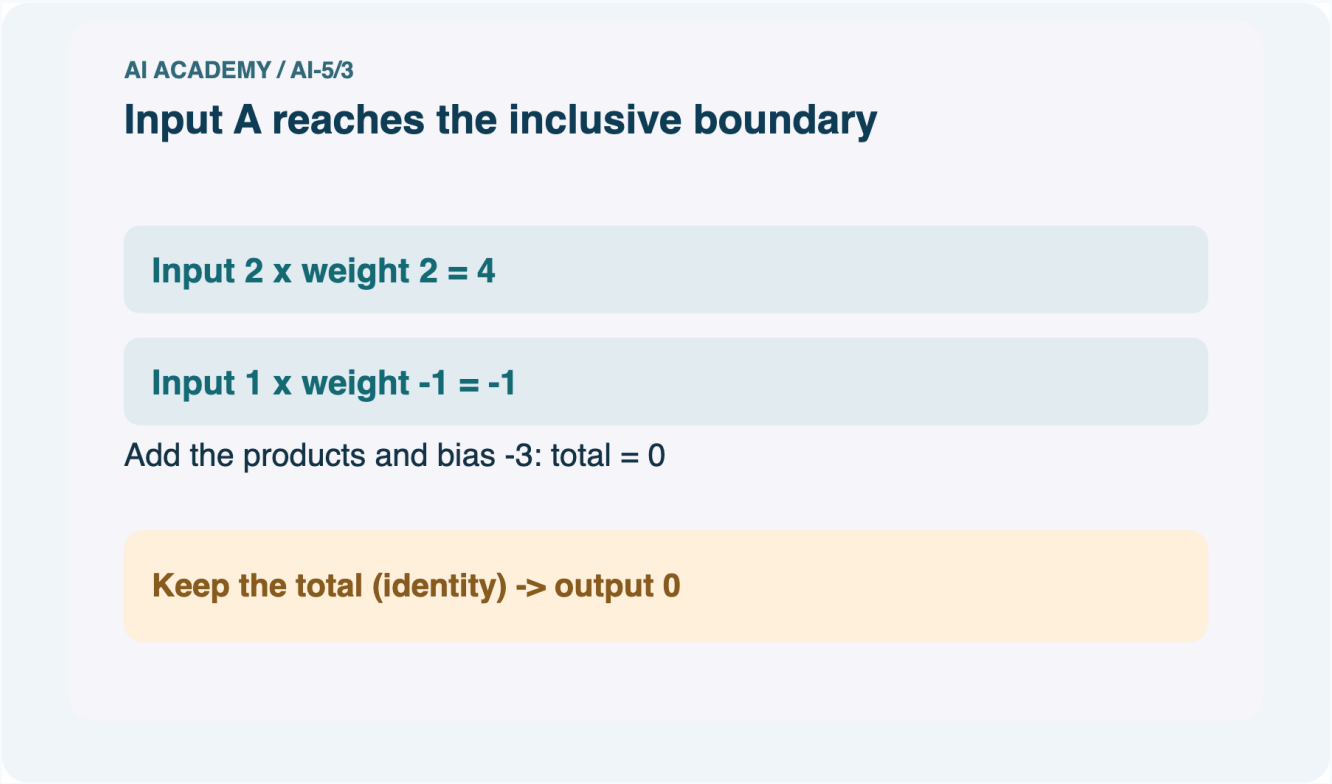
Follow the labelled picture with the matching explanation. Open the vector diagram

Trace **input** A: A score=4-1-3=0, so class 1 at the declared inclusive boundary. → Trace **input** B: B score=2-1-3=-2, so class 0. → Read boundary: The boundary is v=2u-3; the bias sets its position.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Input A reaches the inclusive boundary



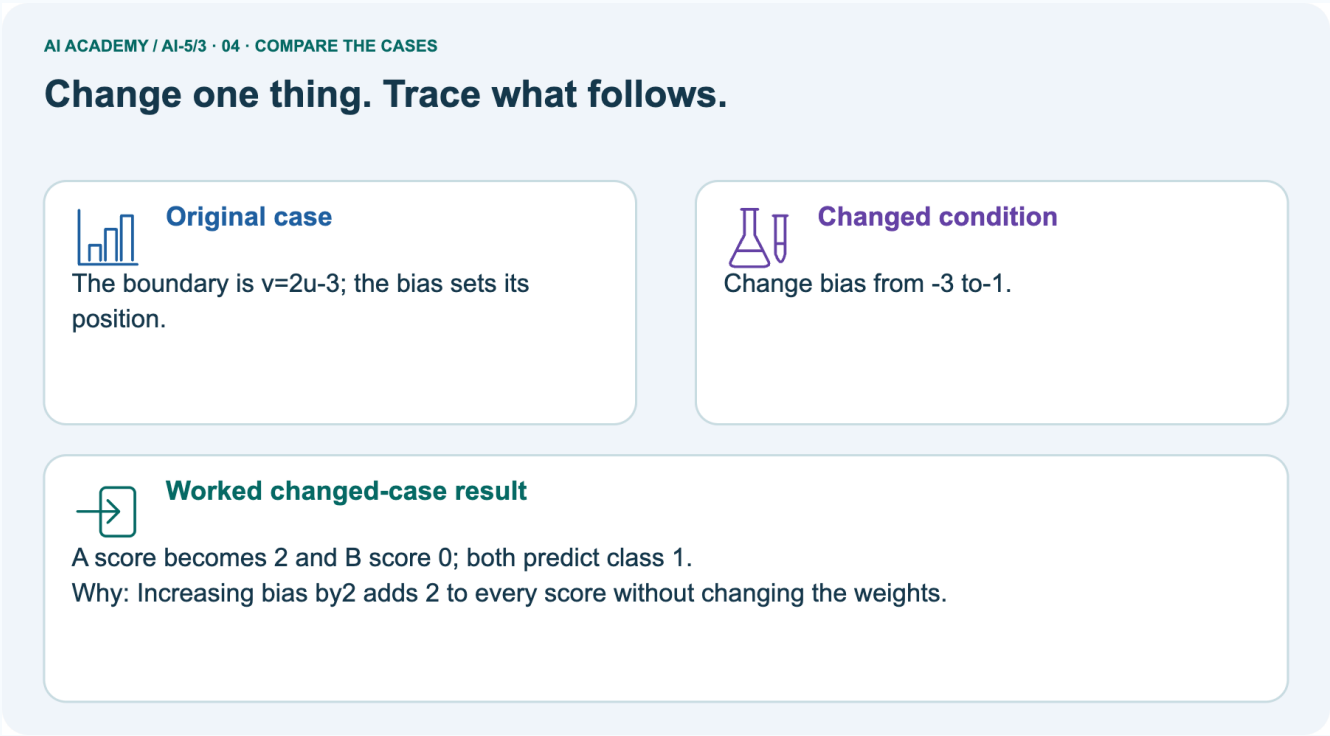
Follow the labelled picture with the matching explanation. Open the vector diagram

Input A reaches the inclusive boundary. **Input**s: [2, 1]; weights: [2, -1]; bias: -3; activation: identity; **output**: 0.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

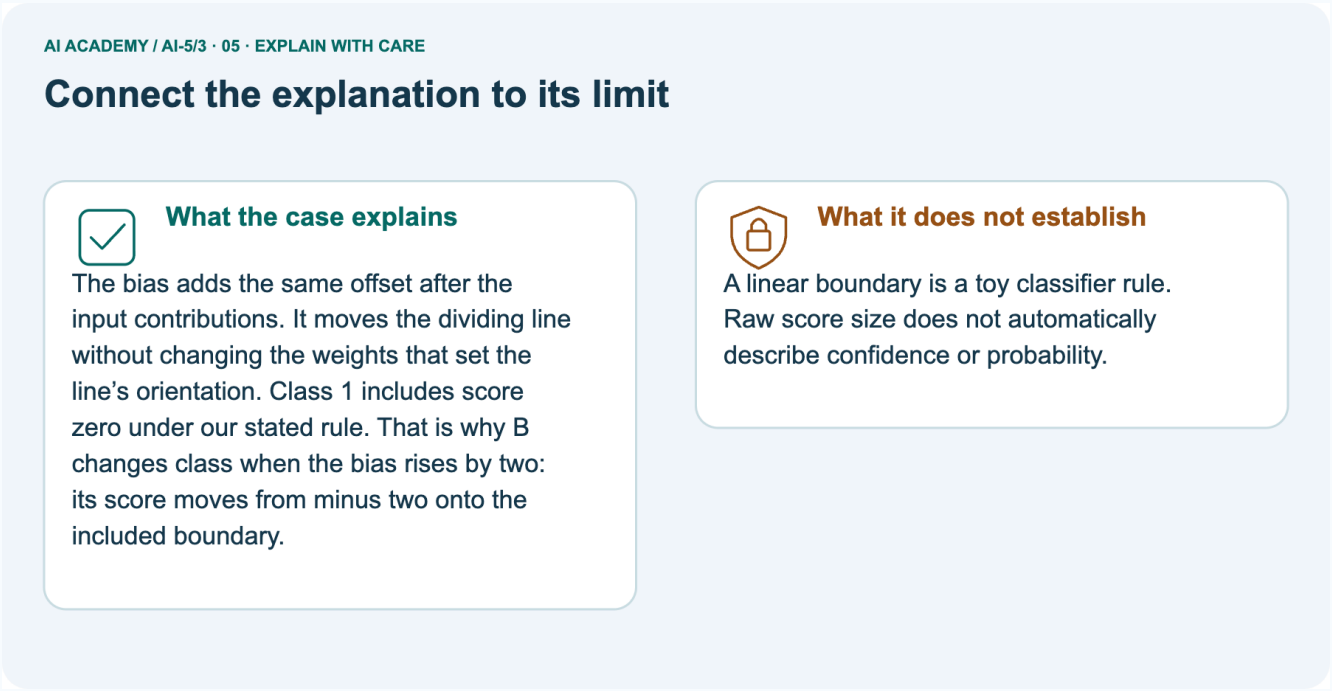
Original case: The boundary is v=2u-3; the bias sets its position. Changed condition: Change bias from -3 to-1. Expected result: A score becomes 2 and B score 0; both predict class 1.

Reason: Increasing bias by2 adds 2 to every score without changing the weights.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The bias adds the same offset after the **input** contributions. It moves the dividing line without changing the weights that set the line's orientation. Class 1 includes score zero under our stated rule.

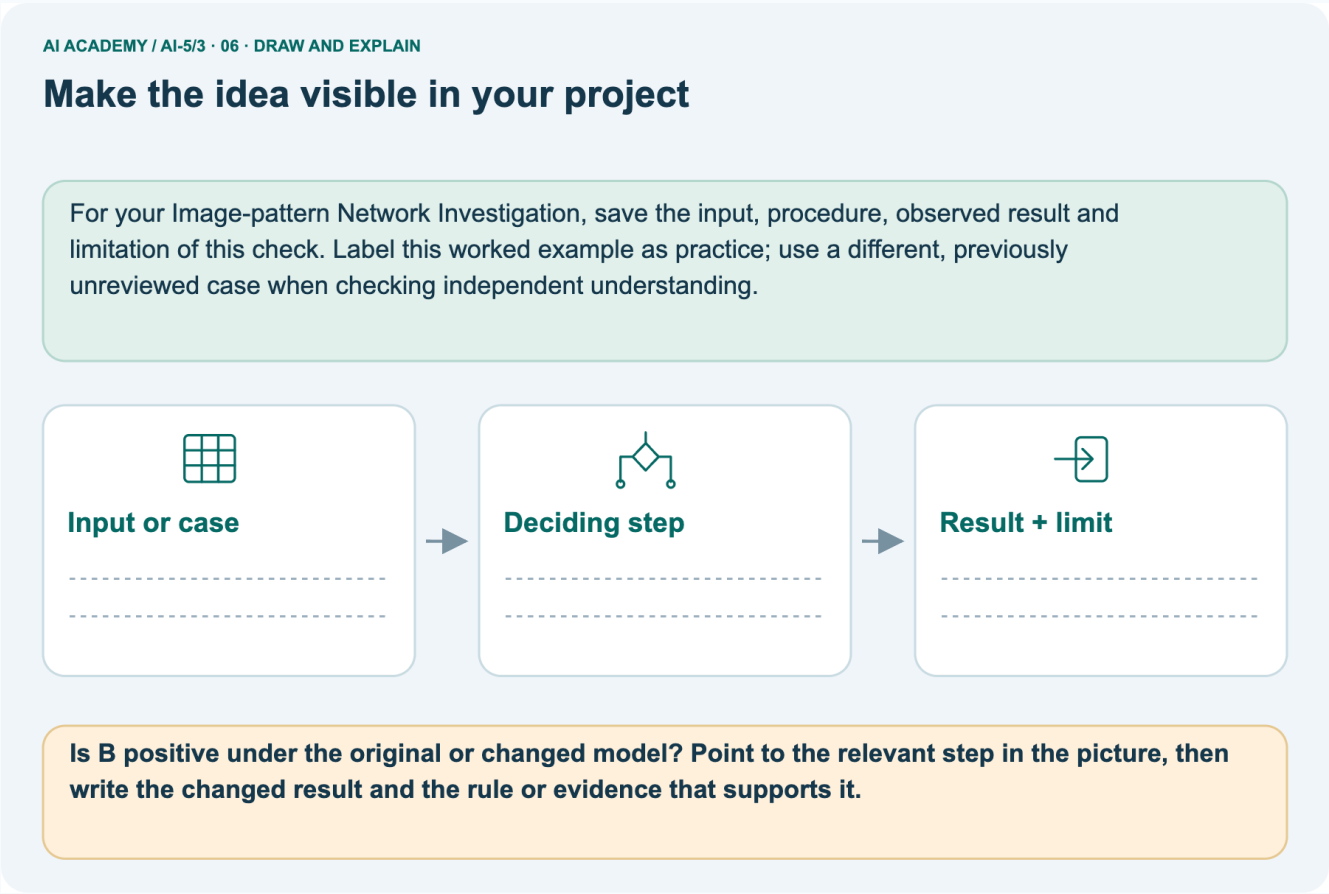
That is why B changes class when the bias rises by two: its score moves from minus two onto the included boundary. Limit: A linear boundary is a toy classifier rule.

Raw score size does not automatically describe confidence or probability.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Is B positive under the original or changed model? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

01. Bias terms and decision boundaries: the case

02. Follow the deciding steps

03. Input A reaches the inclusive boundary

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project
</details>

<details>
<summary>Optional interactive practice and completed work</summary>

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.
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